

PROJECT SYNOPSIS

Secure Orphanage Resource Management System

Database And Management System(UCS310)

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1. INTRODUCTION

The Secure Orphanage Resource Management System is a database-driven platform designed to manage child welfare records and resource distribution in orphanages. Orphanages handle highly sensitive and critical information, including personal details of children, medical history, educational progress, donations received, and allocation of essential resources such as food, clothing, and financial aid.

In many institutions, such information is still maintained through manual registers or loosely structured digital files. These approaches lead to redundancy, inconsistency, limited security, and difficulty in tracking resource utilization. A Database Management System (DBMS) provides a structured, centralized, and secure mechanism for handling such data efficiently.

This project emphasizes backend implementation using SQL and PL/SQL. It incorporates normalization techniques to reduce redundancy, role-based access control to restrict unauthorized access, and encryption mechanisms to ensure confidentiality of sensitive child information. The system aims to ensure data integrity, transparency, and ethical management of orphanage operations.

2. PROBLEM STATEMENT

Orphanages require a structured system to manage welfare records and resources effectively. However, existing manual or file-based systems suffer from the following limitations:

- Data redundancy and inconsistency
- Lack of proper security for confidential child data
- Inefficient tracking of donations and resource allocation
- Absence of structured relationships between entities
- Limited accountability and transparency

There is a need for a secure, well-structured relational database system that ensures confidentiality, maintains referential integrity, and optimizes the allocation of available resources.

The project addresses this gap by developing a normalized database system that securely manages child records, donation tracking, and resource distribution using SQL and PL/SQL-based automation.

3. OBJECTIVES OF THE PROJECT

The major objectives of the project are:

1. To design an Entity–Relationship (ER) model representing orphanage operations.

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2. To convert the ER model into a relational schema with appropriate primary and foreign keys.
 3. To apply normalization up to 3NF/BCNF to eliminate redundancy.
 4. To implement the database using SQL (DDL and DML commands).
 5. To develop advanced SQL queries including joins, subqueries, aggregate functions, and views.
 6. To implement PL/SQL components such as procedures, functions, triggers, and cursors.
 7. To ensure data security using encryption techniques.
 8. To implement role-based access control for Admin, Staff, and Auditor roles.
 9. To ensure transaction consistency using COMMIT, ROLLBACK, and SAVEPOINT commands.

4. SCOPE OF THE PROJECT

The scope of this project is limited to backend database implementation.

Functional Modules:

- Child Welfare Record Management
- Donation Management
- Resource Inventory Management
- Resource Allocation Tracking
- User Role & Access Control

Types of Users:

- **Admin:** Full database privileges
- **Staff:** Data entry and management access
- **Auditor:** Read-only access for transparency

The system focuses strictly on database – level implementation rather than front-end development.

5. PROPOSED SYSTEM DESCRIPTION

The proposed system is a secure relational database implemented using Oracle SQL and PL/SQL. It maintains structured records of children, staff, donations, resources, and allocations.

Key system features include:

- Secure storage of sensitive data using encryption
- Role-based access restrictions
- Automated triggers for audit logging
- Efficient data retrieval using optimized SQL queries
- Transaction management to maintain consistency

The system improves operational efficiency by reducing redundancy and ensuring referential integrity through foreign key constraints. Automated PL/SQL procedures help manage repetitive operations such as allocation processing and reporting.

6. DATABASE DESIGN

6.1 Entity-Relationship (ER) Diagram

Identified Entities:

- Child
- Donation
- Resource
- Staff
- Allocation

Relationships:

- A Child receives Resources
- A Donor provides Donations
- Staff manages Allocations

The ER diagram will define cardinality and participation constraints and will be attached in the final report.

6.2 Relational Schema

Sample relational structure:

- **Child** (Child_ID PK, Name, DOB, Gender, Medical_Record, Education_Details)
- **Donation** (Donation_ID PK, Donor_Name, Amount, Date)
- **Resource** (Resource_ID PK, Resource_Type, Quantity)
- **Allocation** (Allocation_ID PK, Child_ID FK, Resource_ID FK, Date)
- **Staff** (Staff_ID PK, Name, Role, Contact)

Primary and foreign keys ensure referential integrity among tables.

7. NORMALIZATION

The database design will undergo normalization to eliminate redundancy and maintain integrity.

- 1NF: All attributes contain atomic values.
- 2NF: Removal of partial dependencies.
- 3NF: Removal of transitive dependencies.
- BCNF (if applicable): Every determinant is a candidate key.

Functional dependencies will be analyzed before finalizing the schema.

8. DATABASE IMPLEMENTATION

8.1 SQL Implementation

- DDL commands (CREATE, ALTER, DROP)
- DML commands (INSERT, UPDATE, DELETE)
- SELECT queries using joins
- Subqueries
- Aggregate functions
- GROUP BY and HAVING clauses
- Views for reporting

8.2 PL/SQL Components

- Stored Procedures
- Functions
- Triggers
- Cursors
- Exception Handling

9. TRANSACTION MANAGEMENT & CONCURRENCY

Transaction control commands such as COMMIT, ROLLBACK, and SAVEPOINT will be implemented to ensure ACID properties:

- Atomicity
- Consistency
- Isolation
- Durability

Basic concurrency control mechanisms will be discussed to maintain data consistency in multi-user environments.

10. TOOLS & TECHNOLOGIES USED

- Oracle Database
- SQL
- PL/SQL
- SQL developer

11. EXPECTED OUTCOMES

- Successful creation of a normalized database system
 - Secure storage of sensitive child information
 - Efficient resource allocation tracking
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- Automated database operations using PL/SQL
- Improved transparency and accountability in orphanage management

CONCLUSION

The Secure Orphanage Resource Management System demonstrates the practical implementation of core Database Management System concepts in a socially impactful domain. By integrating SQL and PL/SQL-based backend development with normalization, transaction management, and role-based access control, the project ensures secure and efficient management of sensitive orphanage data.

The system enhances transparency, accountability, and proper resource utilization while maintaining strict confidentiality of child information. It also provides valuable practical exposure to real-world database design, implementation, and security techniques. Overall, the project establishes a strong foundation for developing scalable and secure database-driven applications and can be extended into a full-fledged software solution with a user-friendly interface in the future.